Beyond the Hotspot: Spatial and Longitudinal Analysis of Differing Heat Exposure and Warming Trajectories In Bangladesh, 1983-2016

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P9380: Advanced Public Health GIS | Spring 2026

29 April 2026

Introduction

Section I

Public Health Background

Extreme humid heat is one of the most lethal consequences of climate change and Southeast Asia contribute more than half of all global deaths attributed to high temperatures [1]. The problem is spatial because Asia is warming twice as fast as global averages [2].

Research Question

Where is extreme humid heat currently most severe in Bangladesh, and do those areas align with future risk hotspots?

Analytical Approach (Preview)

This project applies a spatial analytical framework to 34 years of satellite data including:

- *Bivariate Local Indicators of Spatial Association (LISA)*
- *Emerging Hotspot Analysis (EHSA)*
- *Bayesian model*

To 64 administrative districts in Bangladesh to examine the spatial and temporal structure of extreme heat risk.

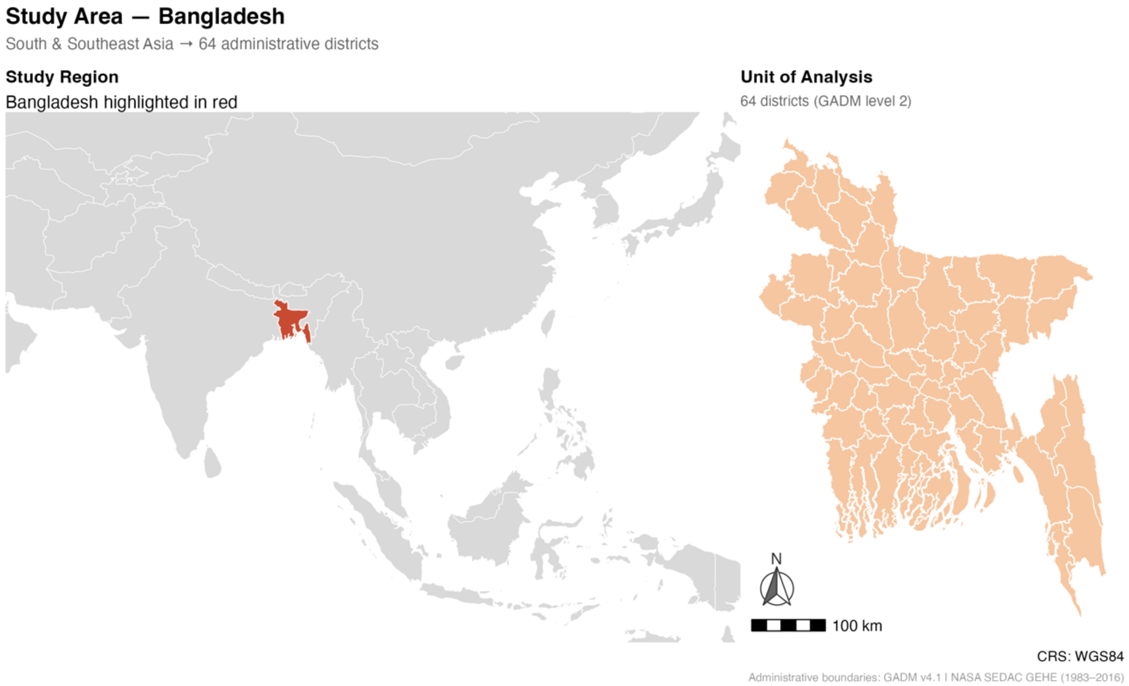


Figure 1. Study Region. [South Asia, Natural Earth]

Figure 2. Study Area. [Bangladesh, GADM boundaries, 2022].

Literature Review

Section II

Theoretical Framework

- Environmental justice framework: *The burden of environmental hazards is not randomly distributed*
- Climate exposure is heterogenous: *Heat risk is shaped by the interaction of physical geography and socioeconomic structure that cannot be shown by one single metric or map*

Key Evidence from the Literature

Heat risk is socially unequal [3]

- *Impacts are worst where adaptive capacity is lowest*

Climate change reinforces inequality [4]

- *Historical warming increased global inequality by ~25%*

South Asia is a global hotspot [5]

- *Region records highest frequency of extreme heat events between 1900 and 2023*

Heat vulnerability is spatially clustered [6]

- *Evidence of high-high clusters in urban heat vulnerability*
- *Spatial autocorrelation detected using Moran's I*

Gap & Contribution

- Prior studies map static heat vulnerability snapshots
- Limited use of longitudinal satellite-based district trends
- Few analyses look at the divergence between current exposure and future warming trajectories across Bangladeshi districts
- Could help determine whether adaptation resources are being directed towards the right places

Methods: Data & Study Area

Section III | Part 1 of 3

Data Source	Variable(s)	Spatial Unit	Year(s)
NASA SEDAC Global Extreme Heat Events (GEHE) *[7] [8]*	Days above WBGTmax28°C threshold	Raster (~4.6km pixel)	1983–2016
Meta / Facebook Relative Wealth Index *[9]*	Relative Wealth Index (RWI) score	Point → aggregated to district	2021
SRTM via NASA (geodata) *[10]*	Mean elevation & Distance to Coast (meters)	Raster (1km) → aggregated to district	2000
GPWv4 NASA SEDAC *[11]*	Population count	Raster (2.5 arc-min) → aggregated to district	2015
GADM v4.1 *[12]*	Administrative Boundaries	Polygon (district, level 2)	2022
Natural Earth *[13]*	Country boundaries	Polygon (country)	2024
World Bank *[14]*	GDP per capita (US)	Country	2019

Spatial Weights:

Bivariate LISA: Queen contiguity via *rgeoda:queen_weights()*

EHSA: Queen contiguity via *sfdep:st_contiguity()*

BYM2: Binary adjacency matrix via *nb2mat()*

Mean Days Above WGBTmax28°C threshold (1983-2016)

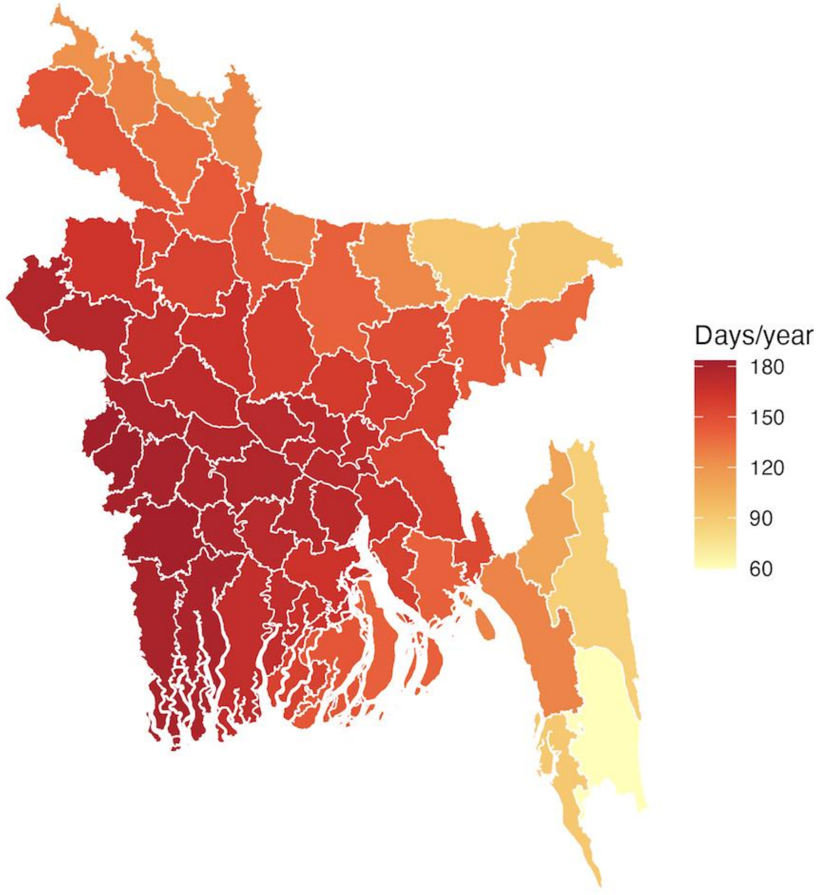


Figure 3. Mean extreme heat exposure by district in Bangladesh.

Data: NASA SEDAC GEHE. Spatial Unit: 64 administrative districts (GADM Level 2)

Methods: Variables

Outcome

Heat Exposure

- *Outcome: WGBT > 28°C days/year (a measure of dangerous, hot-humid conditions)*
- *Distribution: Continuous, ranging from 59.7 to 183.4 days/year*
- *Pixel-level raster values extracted and averaged to district level*

Warming Trend

- *Outcome: change in heat days over time*
- *Definition: OLS slope coefficient from linear regression of annual heat days on time*
- *Distribution: Continuous, ranging from 0.894 to 1.424 days/year*
- *Pixel-level slopes extracted and averaged to district level*

Source: NASA SEDAC Global Extreme Heat Events, 1983-2016

Predictors

Key predictor of interest:

- *Relative wealth index: derived measure of economic status*

Covariates *(all were z-score standardized)*

- *Mean elevation*
- *Distance to coast*
- *Population density*
- *No district-level healthcare access data was available*

Spatial Structure

Spatial Setup

- *Districts (n = 64), Neighbors: Queen contiguity, Weights: Row-standardized (ESDA), binary (BYM2)*

Spatial Structure

- ***Moran's I — Heat Exposure:*** *I = 0.764, p < 2.2e-16, Strong positive spatial autocorrelation*

Moran's I — Warming Trend: *I = 0.716, p < 2.2e-16, Strong positive spatial autocorrelation*

Methods: Analytical Strategy

Section III | Part 3 of 3

Step 1
ESDA

Exploratory Spatial Data Analysis

- **Global Moran's I**

Tested whether heat exposure and warming trends were spatially clustered across Bangladesh districts

Results: I = 0.764 and I = .716, respectively, both p < 2.2e-16

- **Bivariate LISA**

To test whether high heat exposure districts are spatially associated with high warming trend neighbors

Results = Identified 11 low heat/high trend districts

- **Projected crisis timeline analysis**

To identify how many years it would take for 11 low heat-high trend districts to reach high heat numbers using a linear extrapolation

[R: spdep, sfdep, ggplot2]

Step 2
Modeling

Spatial Statistical Model

- **Warming Trend BYM2**

Used to estimate district-level warming rates, accounting for spatial dependence

- **Justification**
- *Bayesian methods capture uncertainty better for small sample n*
- *Adjusts for unmeasured spatial factors*

[R: INLA]

Step 3
Diagnostics

Model Diagnostics & Validation

- **Morans I on residuals**

Used to verify that BYM2 models addressed spatial autocorrelation

- **DIC/WAIC**

Used to evaluate model fit and predictive accuracy without overfitting

[R: spdep, INLA, ggplot2]

Results: Spatial Analysis & Maps

Section IV | ESDA & Spatial Patterns

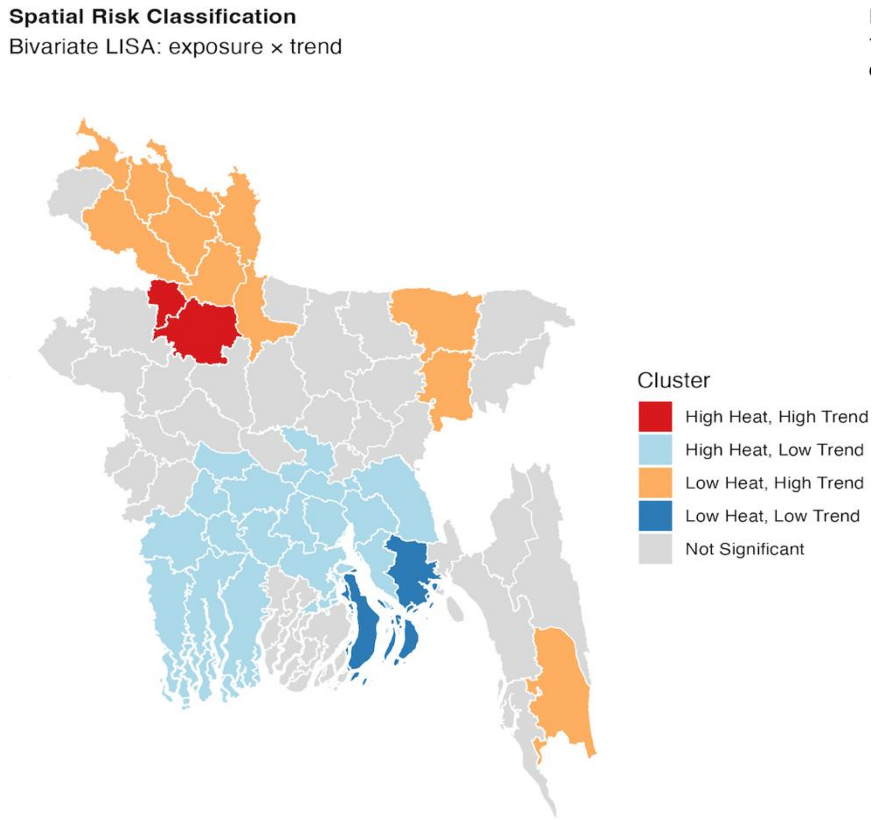


Figure 4. Bivariate LISA – Heat exposure x warming trend. 11 Low heat/high trend districts in the north represent the critical overlooked risk zones. Data: NASA SEDAC GEHE (1983-2016). Created in R.

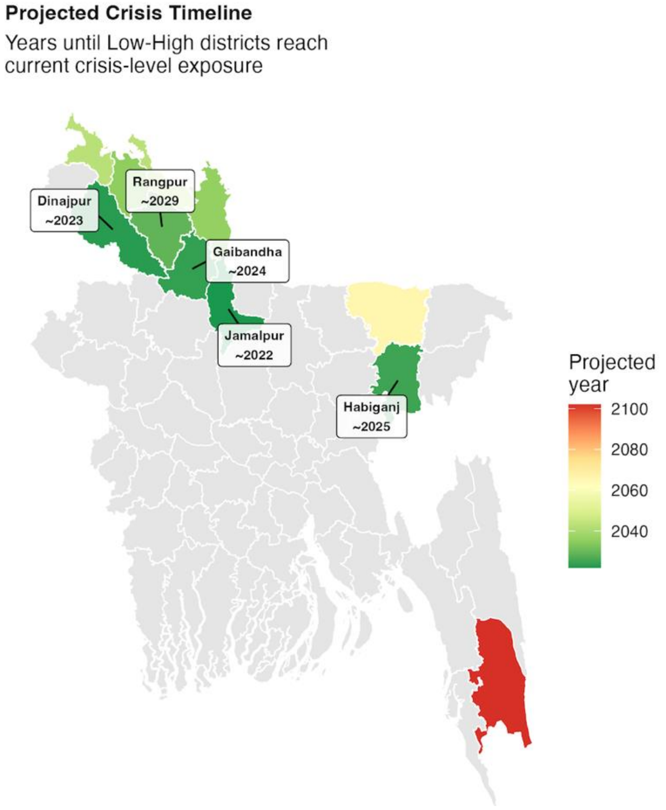


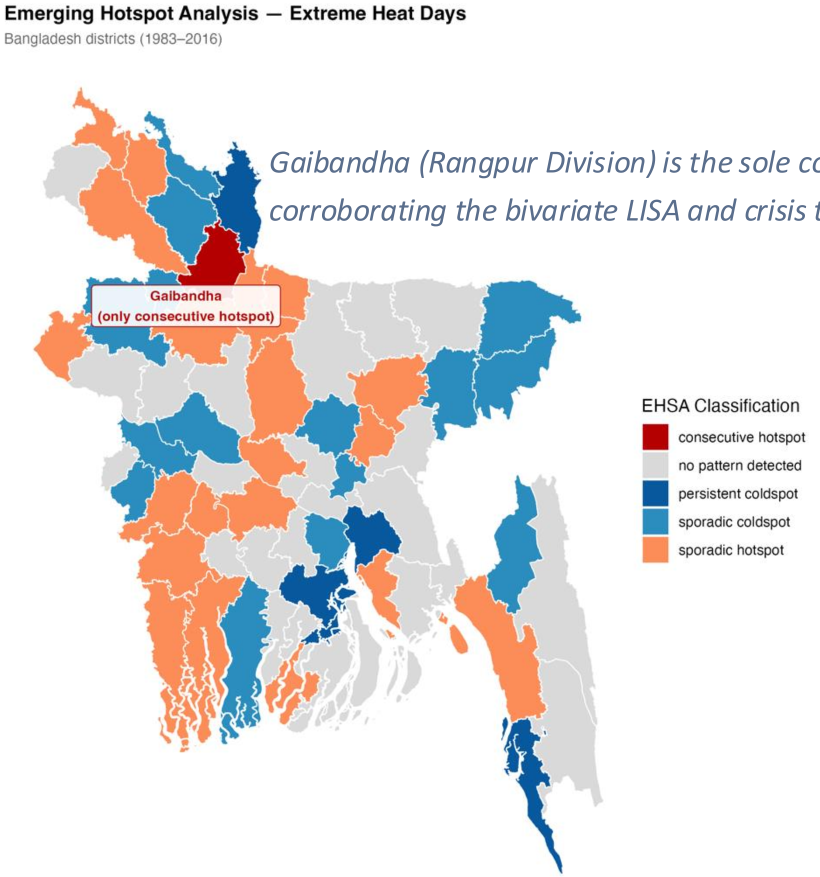
Figure 5. Linear extrapolation of projected heat risk for the 11 low heat/high trend districts identified. Data: NASA SEDAC GEHE (1983-2016). Created in R.

Spatial Statistics: Total districts classified: 32 (50%). High Heat/High Trend: n = 2 (3%). Low Heat/Low Trend: n = 2 (3%).

Low Heat/High Trend: n = 11 (11%). High Heat/Low Trend: n = 17 (27%) . Not significant n = 32 (50%) .

Results: Spatial Analysis & Maps

Section IV | ESDA & Temporal Dynamics



Gaibandha (Rangpur Division) is the sole consecutive hotspot, corroborating the bivariate LISA and crisis timeline findings.

Figure 6. Emerging hotspot analysis . Districts classified by temporal hotspot pattern across 34-year study period (1983-2016). Queen contiguity, k = 1, nsim = 100. Data: NASA SEDAC GEHE (1983-2016). Created in R.

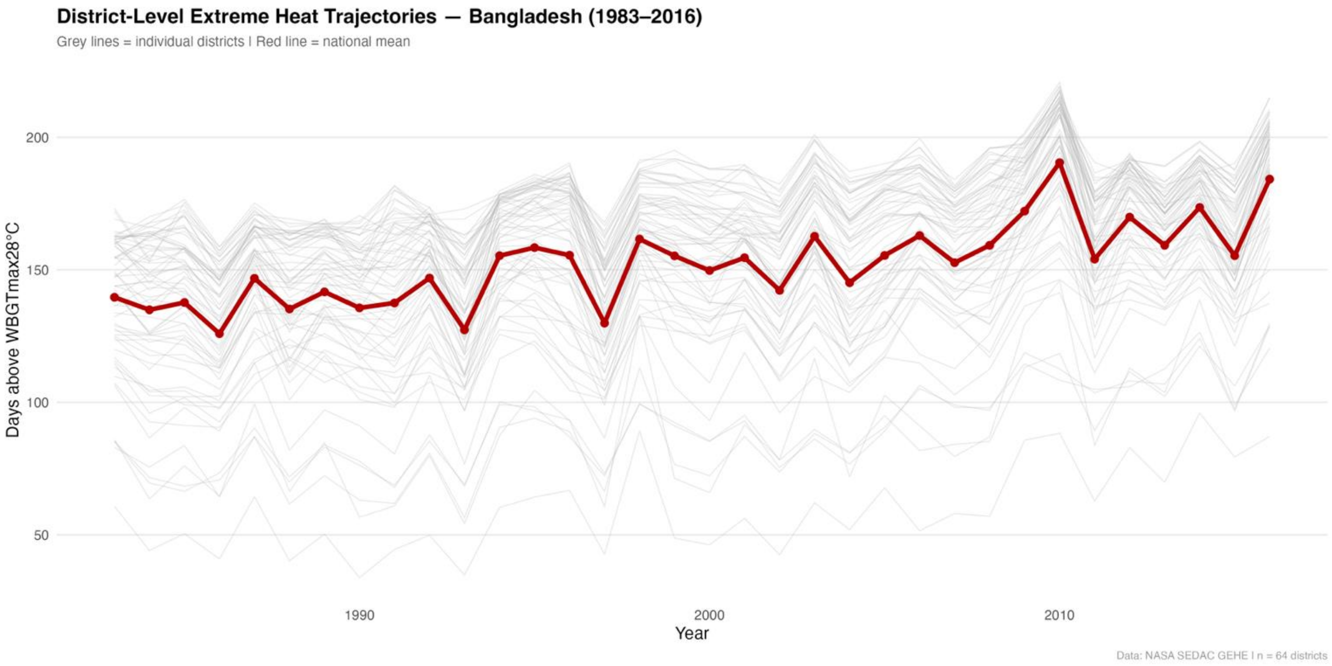


Figure 7. Extreme heat trajectories. Data: NASA SEDAC GEHE (1983-2016). Created in R.

Spatial Statistics: **Consecutive hotspot: n=1 (2%).** Sporadic hotspot: n=20 (31%). Oscillating coldspot: n=19 (30%).
Sporadic coldspot: n=15 (23%). Persistent coldspot: n=4 (6%). No pattern detected: n=5 (8%).

Results: Statistical Models

Section IV | Model Results & Interpretation

Fixed Effects — Cross-Sectional BYM2 Outcome: warming trend (days/year) | Posterior mean (95% credible interval) | * = CI excludes zero[1]	
Variable	**BYM2: Warming Trend**
Intercept	**1.17 (1.165, 1.175)***
Relative Wealth Index	-0.043 (-0.079, -0.007)*
Elevation	-0.043 (-0.072, -0.014)*
Distance to Coast	0.066 (0.011, 0.121)*
Population Density	0 (-0.024, 0.024)

[1] All predictors z-score standardized. * = 95% CI excludes zero.

Hyperparameters and Model Fit Cross-Sectional Bayesian BYM2 via INLA | Gaussian likelihood[1]	
Parameter	**BYM2: Warming Trend**
Phi — spatial mixing (mean)	0.972
Phi — 95% credible interval	(0.873, 0.999)
DIC	-388.28
WAIC	-401.42
Effective parameters	67.97
Moran's I on residuals	-0.052
Residual Moran's I p-value	0.671

[1] Yellow = spatial mixing parameter (97% of residual variation is spatially structured). Green = residual diagnostics (non-significant Moran's I confirms spatial autocorrelation successfully absorbed).

Both tables were created in R (gt)

Key Findings

- *Higher relative wealth index was associated with slower warming (posterior mean = -0.043, 95% CI: -0.079 to -0.007)*
- *The more inland, the faster warming occurs (posterior mean = 0.066, 95% CI: 0.011 to 0.121)*
- *Spatial mixing parameter Phi = 0.972, indicating significant spatial dependence.*
- *Moran's I on residuals = -0.052 (p = 0.671), confirming spatial autocorrelation is addressed.*

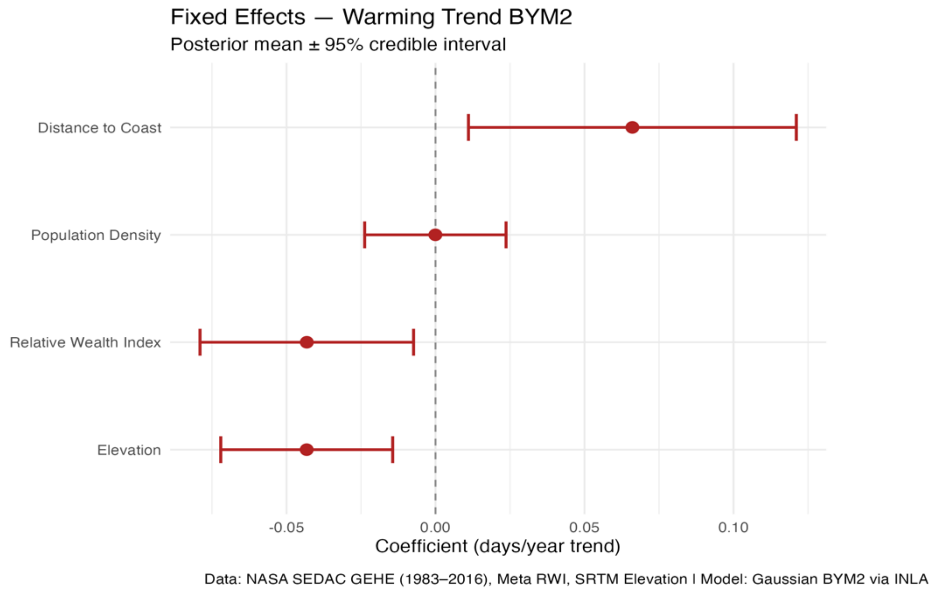


Figure 8. Fixed effects coefficient plots for BYM2 model
Data: NASA SEDAC GEHE (1983-2016) & Meta RWI. Created in R.

Results: Statistical Models

Section IV | What the Model Cannot Explain is Still Spatial

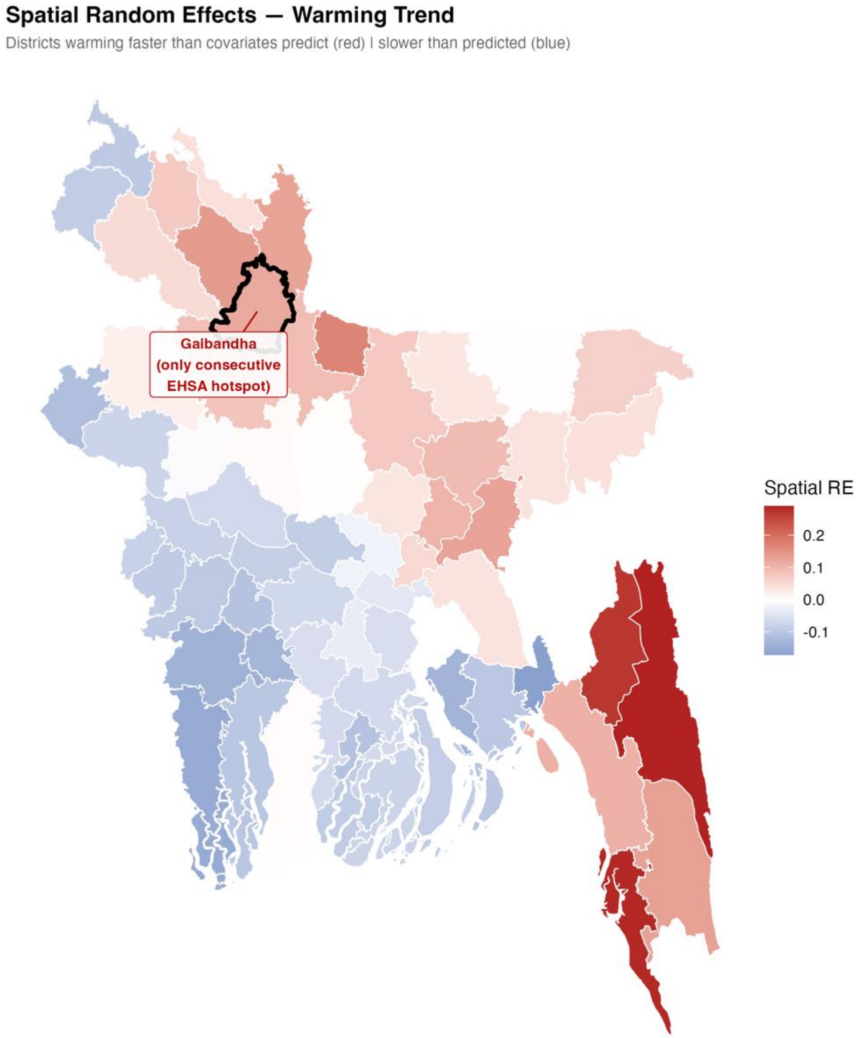


Figure 9. Spatial random effects for warming trend model. Model: Gaussian BYM2 via INLA.
Phi = 0.972 | Residual Moran's I = -0.052 (p = 0.671) Data: NASA SEDAC GEHE (1983-2016) & Meta RWI. Created in R.

Key Findings

- *Gaibandha remains red here, showing that it is warming faster than its wealth, elevation, coastal distance, and population density would predict*
- *Worth noting that the Southeastern districts also show elevated residuals, suggesting that there are additional unmeasured spatial factors influencing this pattern in this region*

Conclusions & Implications

Section V

Finding 1

Spatial analysis reveals that the districts warming the fastest are concentrated in the north. Four independent methods indicate the Gaibandha district as the most critical emerging risk zone.

Finding 2

Wealth drives where heat will be worse in the future, with poorer districts accumulating heat days significantly faster, controlling for other physical characteristics.

Finding 3

Although the current (as of 2016) exposure levels would place the regions of Gaibandha, Jamalpur, and Dinajpur below standard intervention thresholds, they are projected to reach crisis-level heat exposure within a decade of the study period's end.

Public Health Implications

- *Indicates an early intervention window and an opportunity for preventive action before emergencies*
- *Can help with heat-health early warning systems and green infrastructure investments*
- *Key actors: Ministry of Health and Family Welfare Bangladesh and World Health Organization*

Limitations

- *Temporal mismatch among covariates, assuming that socioeconomic and demographic conditions were stable*
- *The linear crisis timeline projection assumes a constant linear warming rate*
- *Small sample size limits the precision of fixed effects coefficients*

Future Directions

This work could be used to create a district-level heat vulnerability index, integrating exposure, sensitivity, and adaptive capacity

References

[1] Cai, J., Maxwell, A. & Zhou, B. Disease burden attributable to high temperature between 1990 and 2021 in South Asia and Southeast Asia, with projections to 2045. (2025). *Trop Med Health* 53(146) https://doi.org/10.1186/s41182-025-00823-4

[2] World Meteorological Organization. (2025). *State of climate in Asia 2024*. https://wmo.int

[3] Romanello, M., Walawender, M., Hsu, S. C., Moskeland, A., Palmeiro-Silva, Y., Scamman, D., Ali, Z., Ameli, N., Angelova, D., Ayeb-Karlsson, S., Basart, S., Beagley, J., Beggs, P. J., Blanco-Villafuerte, L., Cai, W., Callaghan, M., Campbell-Lendrum, D., Chambers, J. D., Chicmana-Zapata, V., Chu, L., … Costello, A. The 2024 report of the Lancet Countdown on health and climate change: facing record-breaking threats from delayed action. (2024). *Lancet (London, England)*, *404*(10465), 1847–1896. https://doi.org/10.1016/S0140-6736(24)01822-1 .

[4] N.S. Diffenbaugh, & M. Burke, Global warming has increased global economic inequality. (2019). Proc. Natl. Acad. Sci. U.S.A. 116 (20) 9808-9813, https://doi.org/10.1073/pnas.1816020116 (2019).

[5] Mani Z, Ranse J, Goniewicz K. Why extreme heat weather is a global health emergency: a retrospective analysis. Int J Emerg Med. (2025) 18(1):255. doi: 10.1186/s12245-025-01062.

[6] Debasish Roy Raja, Md Shah Naim Hredoy, Md. Kamrul Islam, K.M. Ashraful Islam, Mohammed Sarfaraz Gani Adnan, Spatial distribution of heatwave vulnerability in a coastal city of Bangladesh. (2021). Environmental Challenges, Volume 4, 100122, ISSN 2667-0100, https://doi.org/10.1016/j.envc.2021.100122.

[7] Tuholske, C., Peterson, P., Funk, C., & Caylor, K. (2023). *Annual global high-resolution extreme heat estimates (GEHE), 1983–2016* (Version 1.00) [Data set]. NASA Socioeconomic Data and Applications Center (SEDAC). https://doi.org/10.7927/hff0-k565

[8] Tuholske, C., Caylor, K., Funk, C., Verdin, A., Sweeney, S., Grace, K., Peterson, P., & Evans, T. (2021). Global urban population exposure to extreme heat. *Proceedings of the National Academy of Sciences, 118*(41), e2024792118. https://doi.org/10.1073/pnas.2024792118

[9] Chi, G., Fang, H., Chatterjee, S., & Blumenstock, J. E. (2022). Microestimates of wealth for all low- and middle-income countries. *Proceedings of the National Academy of Sciences, 119*(3), e2113658119. https://doi.org/10.1073/pnas.2113658119

[10] NASA JPL. (2013). *NASA shuttle radar topography mission global 1 arc second* (Version 3) [Data set]. NASA Land Processes Distributed Active Archive Center. https://doi.org/10.5067/MEASURES/SRTM/SRTMGL1.003

[11] Center For International Earth Science Information Network-CIESIN-Columbia University. (2017). *Gridded Population of the World, Version 4 (GPWv4): Population Density, Revision 11* (Version 4.11) [Data set]. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). https://doi.org/10.7927/H49C6VHW

[12] GADM. (2022). *Database of global administrative areas, version 4.1* [Data set]. University of California, Berkeley. https://gadm.org

[13] Massicotte P, South A (2026). *rnaturalearth: World Map Data from Natural Earth*. R package version 1.2.0.9000, https://docs.ropensci.org/rnaturalearth/.

[14] Arel-Bundock, V. (2023). *WDI: World Development Indicators (World Bank)*. R package.